CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.2 Transverse & Longitudinal Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

The following statements are about waves.

1	Waves can travel through free space
2	Waves can be polarised
3	Waves have a frequency inversely proportional to their wavelength
4	Waves have an intensity proportional to their amplitude

Which ones would apply to both sound and light waves?

A 3 only

B 4 only

C 1 and 4

D 1 and 2

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The only statement that applies to both light and sound is statement 3, as the relationship between frequency and wavelength is inversely proportional i.e.:
  - $f \propto \frac{1}{\lambda}$
- That means doubling the frequency halves the wavelength

B & **C** are incorrect because the intensity of a wave is proportional to the square of the amplitude

D is incorrect because only transverse waves can be polarised as in a longitudinal wave the particles vibrate in the same direction that the wave travels. Sound waves are not able to travel through free space so statement 1 is incorrect

Question 2 Easy [1 mark]

Ultraviolet waves have an order of magnitude of the frequency of the longest-wavelength of 10^x Hz.

What is the value of *x* ?

A 19

B 17

C 16

D 13

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Ultraviolet waves have a range of frequencies from $7.5 \times 10^{14} – 3 \times 10^{16}$ Hz

Question 3 Easy [1 mark]

Which of the following statements shows that sound waves are longitudinal?

A sound is diffracted around corners

B sound is refracted as it passes from hot air to cold air

C sound cannot be polarised

D sound can be reflected from a solid surface

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Sound is a longitudinal wave
- Longitudinal waves can't be polarised as they oscillate parallel to the direction of travel
- Transverse waves can be polarised as they can oscillate in any plane

A is incorrect because both transverse and longitudinal waves can be diffracted around corners

B is incorrect because both transverse and longitudinal waves can be refracted as they pass from hot to cold air

D is incorrect because both transverse and longitudinal waves can be reflected from a solid surface

Question 4 Easy [1 mark]

Fig 1.1 shows a longitudinal wave.

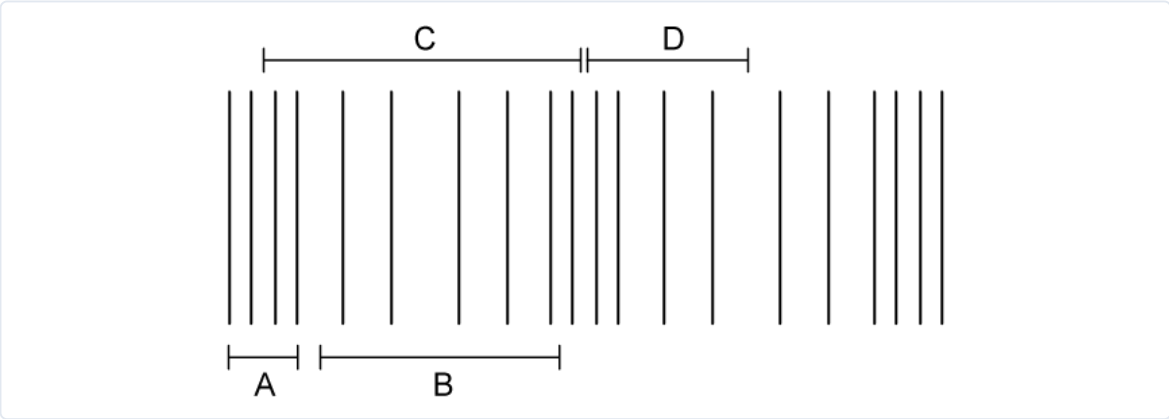


Fig. 1.1

Which label correctly shows the wavelength of a longitudinal wave?

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The label goes from the middle of one compression to another
- This is the wavelength

Option **A** is incorrect because

- The label highlights one compression

Option **B** is incorrect because

- The label highlights one refraction

Option **D** is incorrect because

- The label goes from the middle of a compression to a middle of a refraction
- This is half a wavelength
- For a longitudinal wave, a wavelength can also be measured from one rarefaction to another

Question 5 Easy [1 mark]

Which pair contains one transverse and one longitudinal wave?

A.	Water wave	Visible light
B.	X-ray	Ultrasound
C.	Primary (P) wave	Sound wave
D.	Infrared wave	Vibration on a guitar string

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An X-ray is a transverse wave as it is part of the electromagnetic spectrum
- An ultrasound wave is a longitudinal wave as it is a high frequency sound wave

Option **A** is incorrect because

- Water waves and visible light are both transverse waves

Option **C** is incorrect because

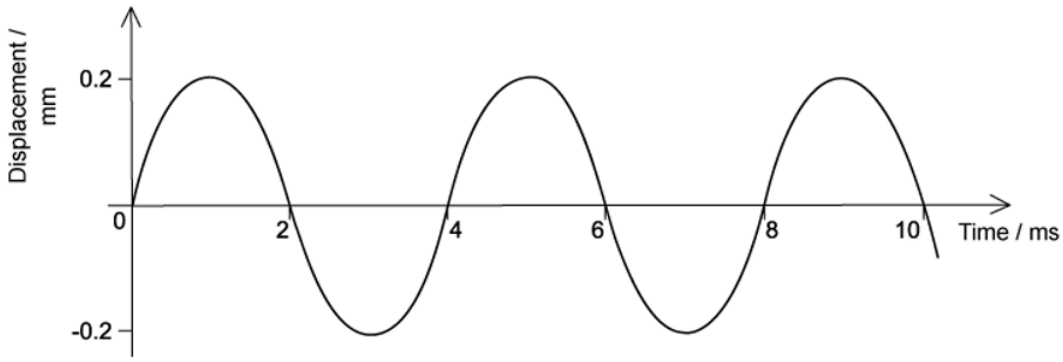
- Primary (P) waves and sound waves are both longitudinal waves

Option **D** is incorrect because

- Infrared waves and vibrations on guitar strings are both transverse waves

Question 6 Medium [1 mark]

Sound travels with a speed of 330 m s^{-1} in air. The variation with time of the displacement of an air particle due to a sound wave is shown below.



Which of the following statements about the wave is correct?

A the wavelength of the sound wave is 1.32 m

B the graph shows that the sound is a transverse wave

C the frequency of the wave is 500 Hz

D the intensity of the wave will be doubled if its amplitude is increased to 0.4 mm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The diagram shows a displacement time graph of a sound wave which is a longitudinal wave
- As time is on the x-axis, we can work out the period but not the wavelength of the wave
- Two wavelengths have a period of $2T$ = 8 ms
  - So, T = 4 ms
- From this frequency can be calculated from the equation
  - $f = \frac{1}{T} = \frac{1}{4 \times 10^{-3}} = 250 \quad \text{Hz}$
- The speed of the wave is calculated by the equation $v = f \quad \lambda$
- Re-arrange for wavelength, $\lambda \quad = \quad \frac{330}{250} = 1.32 \quad \text{m}$

B is incorrect because the graph shows a sine wave or sinusoid wave, this is a graphical depiction of the wave that describes a periodic oscillation this will be seen for both transverse and longitudinal waves

C is incorrect because when the frequency is calculated from the graph it is 250 Hz, not 500 Hz

D is incorrect because the intensity of the wave depends on the square of the amplitude. Doubling of the amplitude multiplies the intensity by a factor of 4

Question 7 Medium [1 mark]

A wave is travelling through a medium at 24 m s^{-1} with a frequency of 15 Hz.

What is the phase difference between two points 2 m apart?

A They are out of phase by half a cycle.

B There is no phase difference.

C They are out of phase by a quarter of a cycle.

D They are out of phase by 0.8 of a cycle.

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Frequency *f* = 15 Hz
- Speed *v* = 24 m s^{-1}
    - The wave equation is $v = f \quad \lambda$
- Wavelength $\lambda = \frac{v}{f} = \frac{24}{15} = 1.6 \quad m$
    - 1.6 m = 1 cycle of the wave
- The question states that there is a phase difference of 2 m so $\frac{2}{1.6} = 1\frac{1}{4}$
- A phase difference that is equal to $1\frac{1}{4}$ is the same as saying it has a phase difference of $\frac{1}{4}$

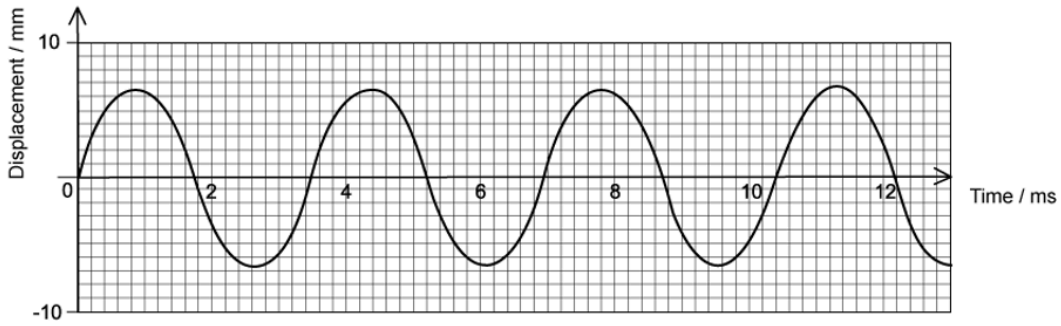
A is incorrect because the distance between two points would be 2.4 m if they were out of phase by half a cycle

B is incorrect because if there was no phase difference then the wavelength and the distance between two points would be equal

D is incorrect because the distance between two points would be 3.0 m if they were out of phase by 0.8 of a cycle

Question 8 Medium [1 mark]

What is the frequency, amplitude and period of the wave shown in the graph below, to 2 significant figures?



	Period / s	Frequency / Hz	Amplitude / m
A	0.0042	240	0.013
B	0.0035	290	0.0065
C	0.0031	320	0.013
D	0.0027	290	0.0065

A

B

C

D

Mark Scheme & Explanation:

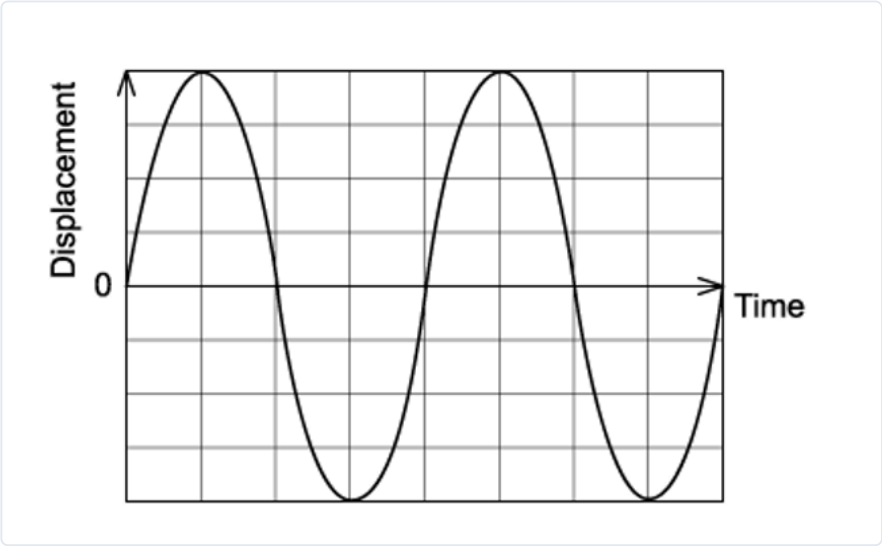
*The correct answer is **B** because:*

- The graph shows displacement against time, where the displacement is given in metres and the time in milliseconds
- It is possible to calculate the period from the graph since there are 3.5 waves in 12.2 milliseconds
  - $3.5 \quad T = 12.2 \times 10^{-3} \quad \text{s}$
  - Period, $T = \frac{12.2 \times 10^{-3}}{3.5} = 0.0035 \quad \text{s}$
  - Frequency, $f = \frac{1}{T} = \frac{1}{0.0035} = 290 \quad \text{Hz}$

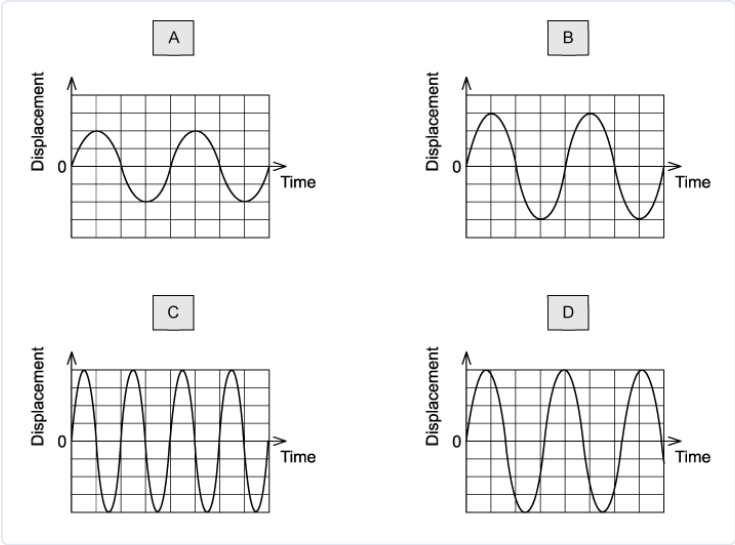
- The amplitude is the maximum displacement of the wave, this can be read from the y axis = 6.5 mm = 0.0065 m

Question 9 Medium [1 mark]

The graph shows the displacement of a sound wave over time.



When the intensity of the wave is halved, which graph shows the correct displacement of the wave ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

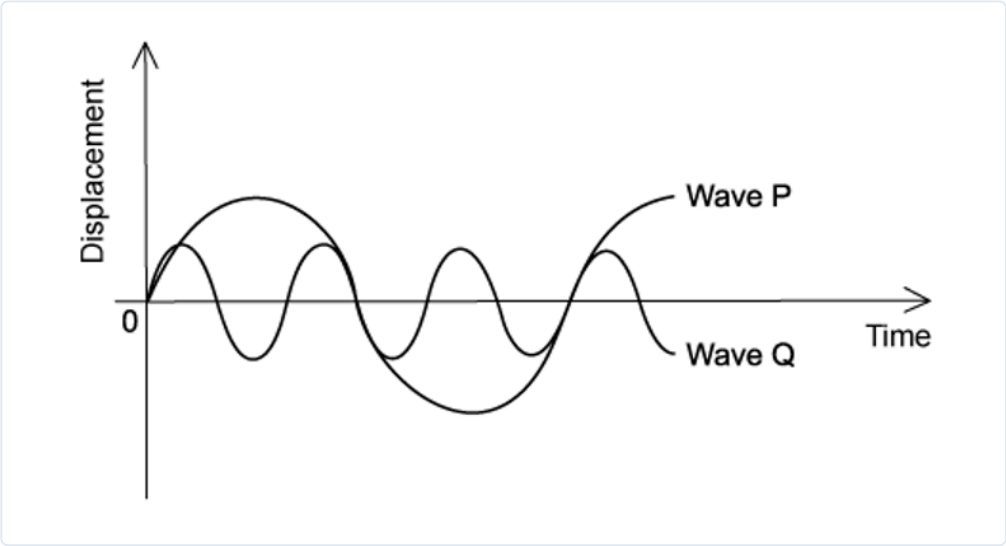
- Intensity *I* is proportional to amplitude squared A^2, i.e.
  - $A^2 \ \alpha \ I \ or \ A \ \alpha \ \sqrt{I}$
- If intensity is halved, $\frac{1}{2}$, then the new amplitude is proportional to $\sqrt{\frac{I}{2}}$
  - $A \ \alpha \ \sqrt{\frac{I}{2}} = \frac{1}{\sqrt{2}}\sqrt{I}$
- The new amplitude is therefore $\frac{1}{\sqrt{2}}A$
  - Therefore, the amplitude is reduced by a factor of $\frac{1}{\sqrt{2}}$ = 0.71 which is shown in graph B

A is incorrect because the graph shows that the amplitude is halved this is not the case as it is reduced by a factor of 0.71

C & **D** are incorrect because the graphs show the same amplitude and increased frequency

Question 10 Hard [1 mark]

The graph shows two waves P and Q.



Wave P has an amplitude of 8 cm and a frequency 100 Hz.

What is the amplitude and the frequency of wave Q?

	Amplitude / cm	Frequency / Hz
A	4	300
B	4	33
C	2	300
D	2	33

A

B

C

D

Mark Scheme & Explanation:

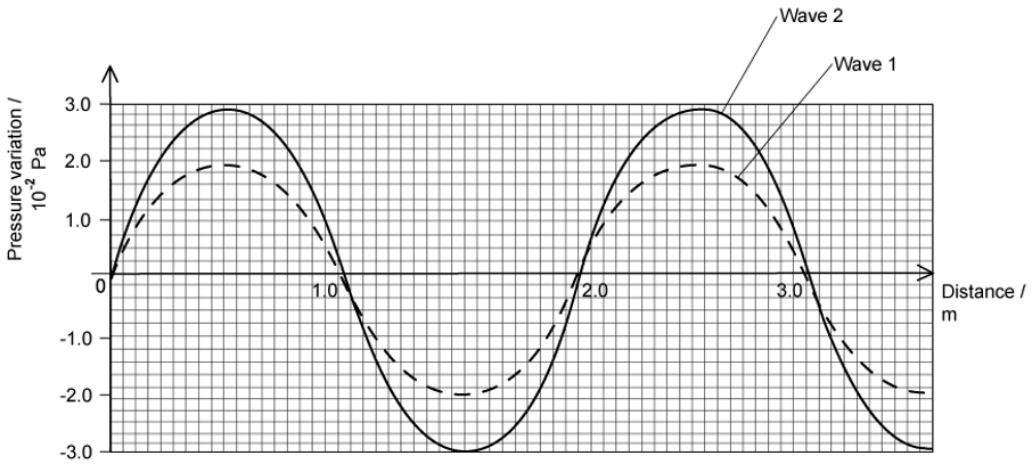
*The correct answer is **A** because:*

- The amplitude is the maximum displacement of the wave
- The amplitude of P is 8 cm and from the diagram we can see wave Q has an amplitude of half that so 4 cm
- Frequency is the number of waves per second of the wave

- Wave P has a frequency of 100 Hz, for every 1 wave that passes of P there are 3 waves of Q
- The period is the time it takes for a wave to complete one wavelength
    - So, the period of P $= \frac{1}{100} = 0.01 \quad \text{s}$
- The period of Q is $= \frac{0.01}{3}$
- The frequency of Q $= \frac{1}{period} = \frac{3}{0.01} = 300 \quad \text{Hz}$
    - Therefore, wave Q has an amplitude of 4 cm and a frequency of 300 Hz

Question 11 Hard [1 mark]

The graph below shows two sound waves, the sound wave is a series of moving pressure variations.



Wave 1 has an intensity of 1.6×10^{-6} W m^{-2}, what is the intensity of wave 2 ?

A 2.4×10^{-6} W m^{-2}

B 3.0×10^{-6} W m^{-2}

C 3.6×10^{-6} W m^{-2}

D 4.5×10^{-6} W m^{-2}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Intensity I is proportional to amplitude squared A^2, i.e.
  - $A^2 \ \alpha \ I$
  - $\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$
- From the graph we can see the amplitude and intensity of wave 1 is equal to
  - $A_1 = 2$
  - $I_1 = 2^2 = 4$
- This corresponds to an intensity of 1.6×10^{-6} W m^{-2}

- From the graph we can see the amplitude and intensity of wave 2 is equal to
  - $A_2 = 3$
  - $I_2 = 3^2 = 9$
- The ratios of I_1 and I_2 can be used to find the corresponding intensity of wave 2
  - $\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$
  - $\frac{1.6 \times 10^{-6}}{I_2} = \frac{4}{9}$
  - $I_2 = \frac{9}{4} \times 1.6 \times 10^{-6} = 3.6 \times 10^{-6}$
- Therefore, the intensity of wave 2 is 3.6×10^{-6} W m^{-2}

Question 12 Hard [1 mark]

Fig 1.1 shows a displacement-distance graph of a transverse wave at t = 0 s. The wave is travelling at 64 m s^{-1}. Particle R is marked at a position of 0.2 m.

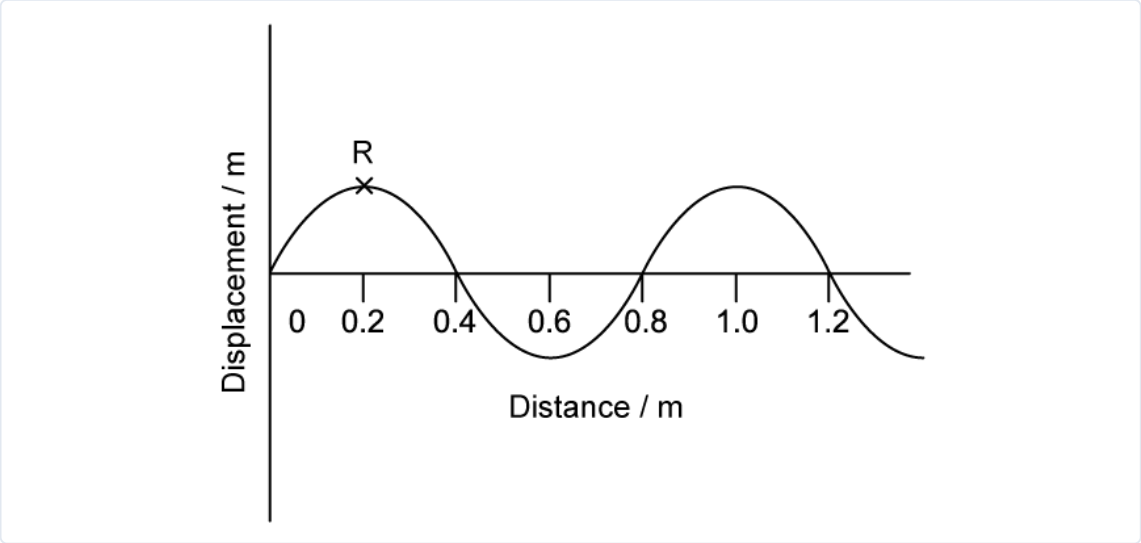


Fig. 1.1

What would be the phase difference between the position of Particle R at t = 0 s and t = 0.034 s?

A 7.0°

B 49°

C 100°

D 260°

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The known quantity is
  - Speed of the wave, v = 64 m s^{-1}
  - Time elapsed, t = 0.034 s
- The wavelength of the wave is the distance from a peak to peak, or trough to trough
  - $\lambda = 1.0 - 0.2 = 0.8$ m
- The wave equation is

- $v = f\lambda$

- The frequency of the wave is therefore
  - $f = \frac{v}{\lambda}$
  - $f = \frac{64}{0.8} = 80$ Hz
- The time period of the wave is
  - $T = \frac{1}{f}$
  - $T = \frac{1}{80} = 0.0125$ s
- The ratio of time *t* and *T* is
  - $\frac{t}{T} = \frac{0.034}{0.0125} = 2.72$
- This means the wave has completed 2.72 oscillations when *t* = 0.034 s
- When a whole oscillation has been completed, the particle will return to its original position
  - The phase difference will therefore be 0.72 of one oscillation
- The phase difference in degrees is
  - $\theta = 0.72 \times 360 = 260°$ (2 s.f)
- This is option **D**

If the question had asked for the phase difference in radians, you would multiply the ratio of $\frac{t}{T}$ by 2π rather than 360°

- $\theta = 0.72 \times 2\pi$ = *4.5 rad (2 s.f)*

Question 13 Hard [1 mark]

A source emits a sound wave and moves away from an observer at a speed of 85 m s^{-1}. Fig 1.1 shows the displacement-distance graph of the sound wave at the source.

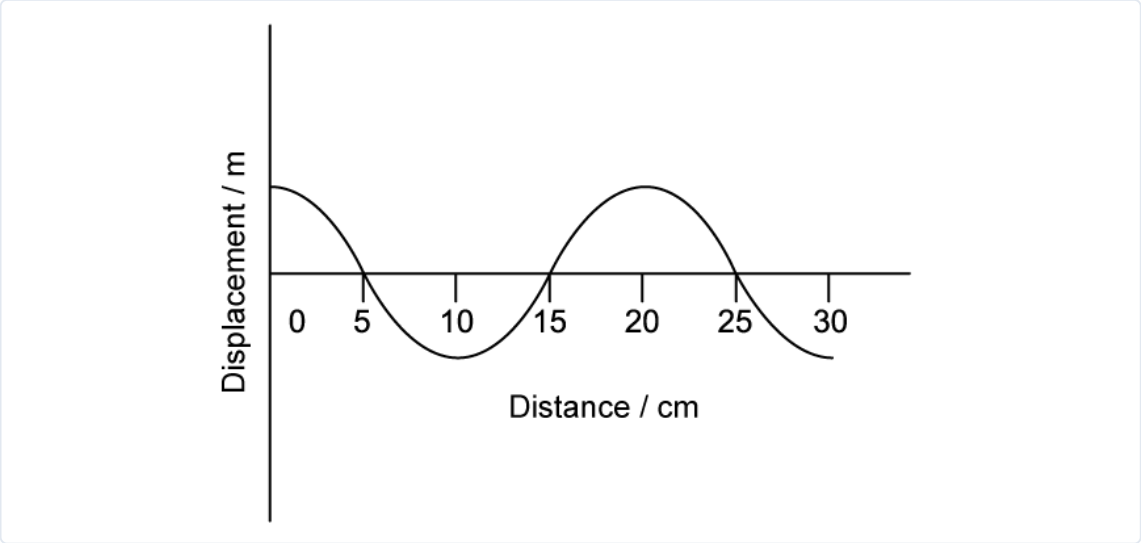


Fig. 1.1

What is the observed frequency of the sound wave?

The speed of sound, v = 340 ms^{-1}

A 1300 Hz

B 1400 Hz

C 1700 Hz

D 2100 Hz

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The known quantities are
  - The speed of sound, v = 340 ms^{-1}
  - The speed of the source, v_s = 85 ms^{-1}
- From the graph, the wavelength of the source is
  - $\lambda = 20 - 0 = 20 \quad \text{cm} = 2 \times 10^{-2} \quad \text{m}$

- The frequency f_s of the sound wave from the source is
  - $v = f_s \lambda$
  - $f_s = \frac{v}{\lambda}$
  - $f_s = \frac{340}{20 \times 10^{-2}} = 1700$ Hz
- The observed frequency of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- When a source is moving away from an observer the denominator is $v + v_s$
- Therefore the observed frequency is
  - $f_o = \frac{1700 \times 340}{340 + 85} = 1400$ Hz (2 s.f)
- This is option **B**

Question 14 Hard [1 mark]

Fig 1.1 shows two longitudinal waves, wave A and wave B, over time *t*. Wave A has an intensity of I_A.

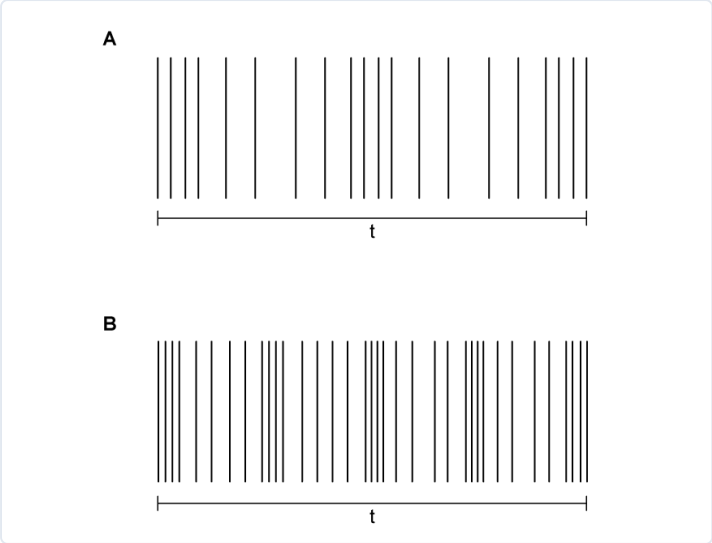


Fig. 1.1

Assuming both waves have equal amplitude, what is the intensity of wave B?

A $\frac{1}{4}I_A$

B $\frac{1}{2}I_A$

C $2I_A$

D $4I_A$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- In wave A, two full waves occur in time *t*
  - This means *t* is twice the period of wave A
  - $t = 2T_A$
- In wave B, four full waves occur in time *t*
  - This means *t* is four times the period of wave B
  - $t = 4T_B$
- The period of wave A is twice the period of wave B

- $T_A = 2T_B$

- Frequency is the reciprocal of period
  - $f_A = \frac{1}{T_A}$
  - Substitute T_B into the above to relate the frequencies of wave A and wave B
  - $f_A = \frac{1}{2T_B} = \frac{1}{2} \times \frac{1}{T_B} = \frac{1}{2}f_B$

- Recall that intensity is proportional to frequency squared
  - $I \propto f^2$

- Therefore the ratio of intensity A to intensity B is
  - $\frac{I_B}{I_A} = \frac{{f_B}^2}{{f_A}^2}$
  - Substitute the expression for frequency A in terms of frequency B
  - $\frac{I_B}{I_A} = \frac{{f_B}^2}{\frac{1}{2}{f_B}^2} = \frac{1}{\frac{1}{2}^2} = 4$

- Rearranging this gives $I_B = 4I_A$